

Software Developer Interview – Simple Explanations (English)

1. React

Definition:

React is a JavaScript library used to build user interfaces (UI) for websites and web applications.

Why use React?

- Builds fast and interactive websites
- Reusable components
- Easy to maintain
- Uses Virtual DOM for better performance

Important Topics

- Components
- Props
- State
- Hooks (`useState` , `useEffect`)
- JSX
- Event Handling
- Conditional Rendering

Example

JavaScript



```
function App() {  
  return <h1>Hello World</h1>;  
}
```

2. Next.js

Definition:

Next.js is a React framework that helps build fast, SEO-friendly, and full-stack web applications.

Why use Next.js?

- Better SEO
- Faster page loading
- Built-in routing
- API creation
- Server-side rendering

Important Topics

- App Router
- Page Router
- Server Components
- Client Components
- API Routes
- Dynamic Routing
- Middleware
- Image Optimization

3. PostgreSQL

Definition:

PostgreSQL is an open-source relational database used to store application data.

Example Data

Users



ID

Name

Email

Password

Why use PostgreSQL?

- Secure
- Fast
- Reliable
- Supports large applications

Important Topics

- Tables
- Rows
- Columns
- Primary Key
- Foreign Key

- SQL Queries
 - Relationships
 - Indexes
-

4. Prisma ORM

Definition:

Prisma is an ORM (Object Relational Mapper) that allows developers to communicate with a database using JavaScript or TypeScript instead of writing SQL.

Instead of

SQL

```
SELECT * FROM users;
```



You write

TypeScript

```
await prisma.user.findMany();
```



Why use Prisma?

- Easy database operations
- Type-safe queries
- Auto-generated client
- Database migrations

Important Topics

- Prisma Schema
 - Models
 - Migrations
 - Prisma Client
 - CRUD Operations
 - Relations
-

5. REST API

Definition:

REST API allows the frontend and backend to communicate using HTTP requests.

Example

```
Frontend
  ↓
REST API
  ↓
Database
```



HTTP Methods

- GET → Read data
- POST → Create data
- PUT → Update all data
- PATCH → Update part of data
- DELETE → Delete data

Important Topics

- Request
- Response
- Status Codes
- JSON
- Endpoints
- Authentication

6. Solidity & Ethereum

Solidity

Solidity is a programming language used to create Smart Contracts.

Example

```
Bank Contract
Voting Contract
NFT Contract
```



Ethereum

Ethereum is a blockchain platform where Smart Contracts are deployed.

Important Topics

- Smart Contracts
- Blockchain

- Gas Fee
 - Wallet
 - MetaMask
 - Transactions
 - ETH
-

7. App Router & Page Router

Page Router (Old)

pages/

index.js
about.js
contact.js



Uses

- `getServerSideProps()`
 - `getStaticProps()`
-

App Router (New)

app/

page.tsx
layout.tsx
loading.tsx
error.tsx



Advantages

- Better performance
- Nested layouts
- Server Components
- Easier routing

Important Topics

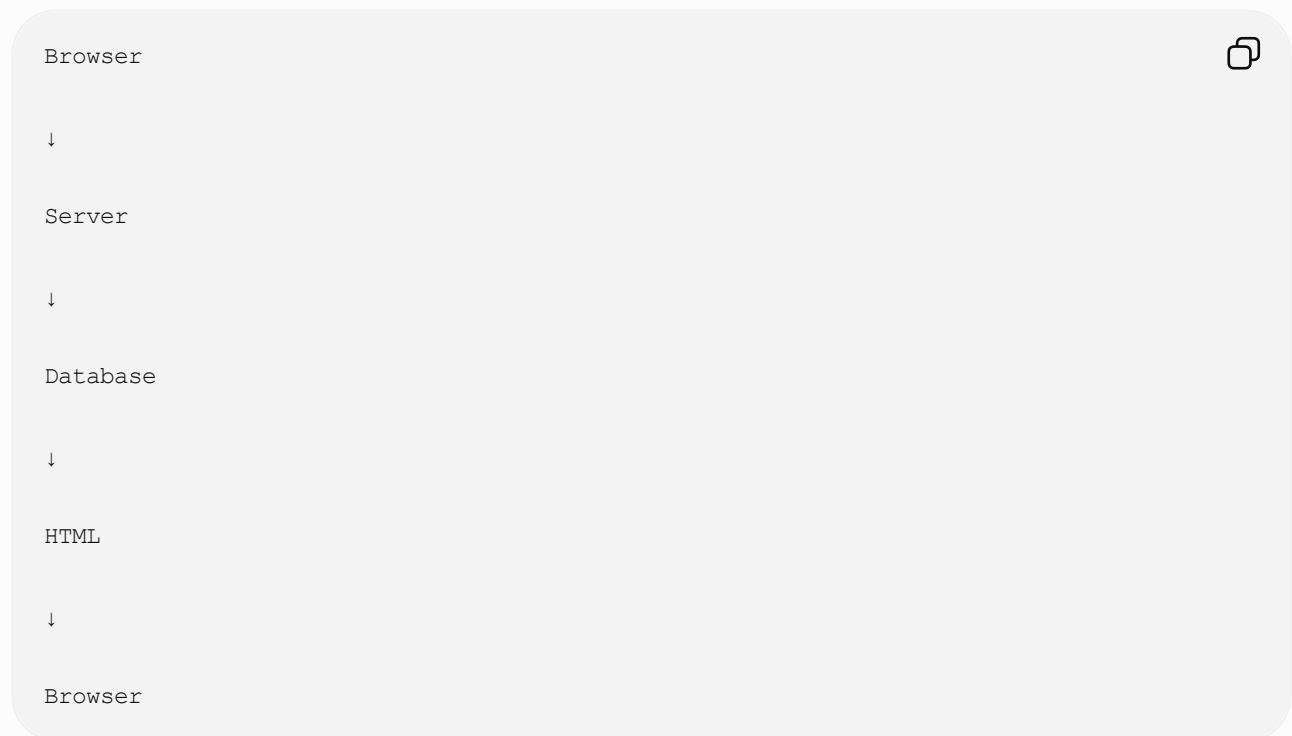
- `page.tsx`
- `layout.tsx`
- `loading.tsx`
- `error.tsx`

- Route Groups
- Dynamic Routes

8. SSR (Server Side Rendering)

Definition

The page is rendered on the server before being sent to the browser.



Advantages

- Faster first page load
- Better SEO
- Better user experience

Other Rendering Types

- SSR
- CSR (Client Side Rendering)
- SSG (Static Site Generation)
- ISR (Incremental Static Regeneration)

9. TypeScript

Definition

TypeScript is JavaScript with type checking.

JavaScript

JavaScript



```
let age = 25;
```

TypeScript

TypeScript



```
let age: number = 25;
```

Advantages

- Finds errors early
- Better IntelliSense
- Easier to maintain
- Safer code

Important Topics

- Types
- Interfaces
- Enums
- Generics
- Type Aliases
- Functions
- Objects

10. Relational Data Modeling

Definition

Relational Data Modeling is the process of organizing data into related tables.

Example

Users



ID

Name

↓

Projects

ID
Title
UserID

Relationships

- One-to-One
- One-to-Many
- Many-to-Many

Important Topics

- Primary Key
- Foreign Key
- Normalization
- Relationships
- Constraints

11. JWT

Definition

JWT (JSON Web Token) is a secure token used to verify a user's identity after login.

Flow

Login



↓

JWT Created

↓

Browser Stores Token

↓

Token Sent with Requests

↓

Server Verifies

↓

Access Granted

Important Topics

- Access Token
- Token Verification
- Expiration Time
- Authorization

12. bcryptjs

Definition

bcryptjs encrypts (hashes) passwords before saving them in the database.

Example

Password



↓

Hashed Password

↓

Database

Why use it?

- Passwords are stored securely.
- Even if the database is leaked, the original password is difficult to recover.

Important Topics

- Hashing
- Salt
- Compare Password

13. Zod Validation

Definition

Zod validates user input before processing or saving it.

Example

Email Required



Benefits

- Prevents invalid data
- Better error messages
- Type-safe validation

Important Topics

- Schema
- Validation
- Error Handling
- Type Inference

14. Cookie-Based Authentication

Definition

After login, the server stores a secure cookie in the user's browser. The browser automatically sends the cookie with future requests.

Flow

Login



↓

Cookie Created

↓

Browser Stores Cookie

↓

Browser Sends Cookie

↓

Server Verifies

↓

User Authenticated

Advantages

- More secure with `HttpOnly` cookies
- Automatic authentication

15. Protected Routes

Definition

Protected Routes are pages that only logged-in users can access.

Example

Dashboard

Profile

Admin Panel



If the user is not logged in:

Redirect

↓

Login Page



Important Topics

- Authentication
- Authorization
- Route Protection
- Middleware

16. Git

Definition

Git is a version control system used to track code changes.

Common Commands

❏ Bash

```
git init
```

```
git add .
```



```
git commit -m "First Commit"
git push
git pull
git clone
```

Important Topics

- Repository
 - Commit
 - Branch
 - Merge
 - Push
 - Pull
 - Clone
-

17. GitHub

Definition

GitHub is an online platform used to store and manage Git repositories.

Uses

- Store code online
 - Team collaboration
 - Pull Requests
 - Issue Tracking
 - GitHub Actions
-

18. GitLab

Definition

GitLab is another platform for hosting Git repositories with built-in DevOps and CI/CD features.

Uses

- Source code management
 - CI/CD pipelines
 - Merge Requests
 - Issue Tracking
-

19. VS Code

Definition

Visual Studio Code (VS Code) is a popular code editor used to write and debug code.

Features

- IntelliSense
- Extensions
- Terminal
- Git Integration
- Debugger

Important Topics

- Extensions
 - Terminal
 - Debugging
 - Source Control
-

20. Postman

Definition

Postman is a tool used to test and debug APIs.

What can you do?

- Send GET, POST, PUT, DELETE requests
- Test APIs
- Check responses
- Save collections

Important Topics

- HTTP Methods
 - Headers
 - Body
 - Authorization
 - Environment Variables
-

21. Vercel

Definition

Vercel is a cloud platform used to deploy web applications, especially Next.js projects.

Features

- One-click deployment
- Automatic HTTPS
- Preview deployments
- Serverless Functions
- Custom Domains

Important Topics

- Deployment
- Environment Variables
- Build Process
- Preview URLs
- Production Deployment

22. CI/CD Deployment Workflow

Definition

CI/CD is an automated process that builds, tests, and deploys applications whenever code changes are pushed.

Workflow

Developer



↓

Write Code

↓

Git Commit

↓

GitHub / GitLab

↓

CI Pipeline

```
graph TD; A[ ] --> B[Build]; B --> C[Test]; C --> D[Deploy]; D --> E[Vercel / Server]; E --> F[Live Website];
```

↓

Build

↓

Test

↓

Deploy

↓

Vercel / Server

↓

Live Website

CI (Continuous Integration)

- Automatically builds and tests code after every commit.

CD (Continuous Deployment/Delivery)

- Automatically deploys successful builds to production or a staging environment.

Benefits

- Faster deployments
- Fewer manual errors
- Consistent releases
- Easier team collaboration

Important Topics

- Build
- Test
- Deploy
- Pipelines
- Automation
- Environment Variables

These explanations are suitable for beginners and also cover the core concepts commonly asked in React, Next.js, and Full Stack Developer interviews.